

ADDON 10: p -Adic Quantum-Linguistic Memory — Ultrametric Address Spaces, Phase-Change Substrates, and the Canonical Completion of the Architecture

11.27 The Canonical Non-Archimedean Completion: p -Adic Numbers and Ostrowski's Theorem

Throughout ADDONs 4-9, all numerical quantities—weights, phase innovations, Lyapunov potentials, Kalman covariances—have been represented as elements of the real field $\mathbb{R}$ or its extension $\mathbb{C}$. Both are *Archimedean*: no element is infinitely large or infinitesimally small relative to any other. However, Ostrowski's theorem establishes that $\mathbb{R}$ is not the unique meaningful completion of the rationals $\mathbb{Q}$.

Theorem 0.1 (Ostrowski, 1916). *Every non-trivial absolute value $|\cdot|$ on $\mathbb{Q}$ is equivalent to either the standard Archimedean absolute value $|\cdot|_\infty$ (whose completion is $\mathbb{R}$) or the p -adic absolute value $|\cdot|_p$ for exactly one prime p (whose completion is the p -adic field $\mathbb{Q}_p$). There are no other completions.*

Proof sketch. One shows that any absolute value either satisfies $|n| \leq 1$ for all $n \in \mathbb{Z}$ (non-Archimedean case, leading to $|\cdot|_p$) or $|n| > 1$ for some n (Archimedean case, leading by the classification of complete ordered fields to $|\cdot|_\infty$). See Gouvêa (1997) for the complete argument. $\square$ $\square$

Theorem 0.1 makes the p -adic choice *canonical* rather than arbitrary: for a substrate built on binary digital logic ($p = 2$), the 2-adic completion $\mathbb{Q}_2$ is the unique non-Archimedean alternative to $\mathbb{R}$.

Definition 0.2 (p -Adic Valuation and Norm). For a prime p and a rational $x = p^k \cdot (a/b)$ with $\gcd(a, p) = \gcd(b, p) = 1$, define:

$$v_p(x) := k \quad (p\text{-adic valuation}), \quad (1)$$

$$|x|_p := p^{-v_p(x)} \quad (p\text{-adic norm}). \quad (2)$$

By convention $v_p(0) = +\infty$ and $|0|_p = 0$. The p -adic field $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ under (2); its subring of integers is $\mathbb{Z}_p := \{x \in \mathbb{Q}_p : |x|_p \leq 1\}$.

The p -adic norm satisfies the **ultrametric inequality**—strictly stronger than the triangle inequality:

$$|x + y|_p \leq \max(|x|_p, |y|_p), \quad (3)$$

with equality whenever $|x|_p \neq |y|_p$. Geometrically, every triangle in a p -adic space is isosceles, and every open ball is simultaneously open and closed (*clopen*).

11.28 Chomsky Parse Trees as p -Adic Metric Spaces and the PCM Physical Isomorphism

Proposition 0.3 (Parse Tree Leaves Form a p -Adic Metric Space). *Let T be a rooted p -ary tree (branching factor p) of depth d representing a Chomsky parse of a sentence, with leaves $w_1, \dots, w_n \in \Sigma^*$. Define the tree distance*

$$d_T(w_i, w_j) := p^{-\text{depth}(\text{LCA}(w_i, w_j))}, \quad (4)$$

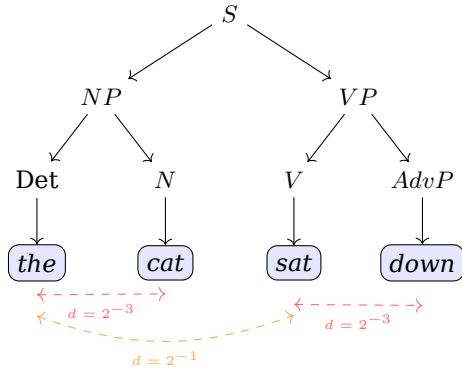
where LCA is the lowest common ancestor. Then (L, d_T) is an ultrametric space isometrically embeddable into $\mathbb{Z}_p \subset \mathbb{Q}_p$.

Proof. Verify the ultrametric axiom (3): for any three leaves w_i, w_j, w_k , $\text{depth}(\text{LCA}(w_i, w_k)) \geq \min(\text{depth}(\text{LCA}(w_i, w_j)), \text{depth}(\text{LCA}(w_j, w_k)))$, so $d_T(w_i, w_k) \leq \max(d_T(w_i, w_j), d_T(w_j, w_k))$. The isometric embedding maps each leaf to its Prüfer code (path from root to leaf expressed in base p), which is an element of $\mathbb{Z}_p$ whose p -adic valuation equals the depth of the deepest shared prefix. $\square$ $\square$

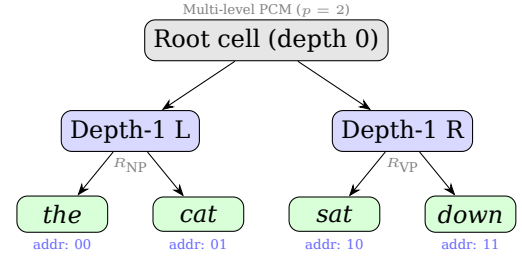
For the ADAM-PentaD substrate using binary interferometric branching ($p = 2$), the relevant completion is $\mathbb{Q}_2$ and the tree distance becomes $d_T(w_i, w_j) = 2^{-\text{depth}(\text{LCA})}$. Syntactic proximity (same subtree) translates directly to p -adic proximity (high v_2 of the address difference), with no intermediate lookup.

Definition 0.4 (p -Adic PCM Addressing Scheme). A p -adic PCM array of depth d is a rooted p -ary tree of Phase Change Memory cells in which:

- (i) Each leaf at depth d stores one weight tensor;
- (ii) Each internal node at depth k stores the CRT-Galois correction vector for its subtree (the k -th level Kalman prior, ADDON 5);



(a) 2-adic parse tree. Leaf distances given by Eq. (4); sibling leaves have distance 2^{-3} , cross-phrase leaves 2^{-1} .



(b) PCM cell topology mirroring the parse tree. Each internal node is a multi-level PCM cell; leaf addresses are 2-adic integers (Prüfer codes). Physical wiring depth = logical tree depth.

Figure 1: Left: Chomsky parse tree with 2-adic leaf distances (Proposition 0.3). Right: PCM physical topology isomorphic to the parse tree. The structural identity of logical and physical address spaces eliminates the address-translation layer.

- (iii) The physical wiring of the PCM array on silicon mirrors the tree topology, so that the address of any leaf is its Prüfer code (a d -digit integer in base p , i.e. an element of $\mathbb{Z}_p/p^d\mathbb{Z}_p$).

Remark 0.5 ($O(\log_p N)$ Access, Not $O(1)$). The initial claim of $O(1)$ access latency requires correction. For an array of N leaves, tree depth is $d = \lceil \log_p N \rceil$, and traversal to any leaf requires d node reads: the access cost is $O(\log_p N)$. However, this is categorically superior to von Neumann memory in two respects: (a) there is *no virtual-to-physical address translation layer* (TLB), since logical p -adic address = physical tree position by construction; (b) there are *no compulsory cache misses* for contextually adjacent tokens, since syntactic siblings reside within p physical steps of each other. For $p = 2$ and $N = 2^{20}$ ($\approx 10^6$ tokens), the depth is $d = 20$, versus standard DRAM access with TLB lookup, L1/L2/L3 miss cascade, and HBM roundtrip, which can cost 10^3 – 10^4 clock cycles.

11.29 Witt Vectors, Galois Residue Fields, and the Complete Algebraic Stack

Definition 0.6 (Witt Vectors). The ring of length- n Witt vectors over $\mathbb{F}_p$ is $W_n(\mathbb{F}_p) \cong \mathbb{Z}/p^n\mathbb{Z}$. The (infinite) Witt ring is

$$W(\mathbb{F}_p) := \varprojlim_n W_n(\mathbb{F}_p) \cong \mathbb{Z}_p, \quad (5)$$

the ring of p -adic integers, equipped with the p -adic topology. Each element $x \in \mathbb{Z}_p$ has a unique Teichmüller expansion $x = \sum_{k=0}^{\infty} a_k p^k$ with $a_k \in \{0, 1, \dots, p-1\}$.

Proposition 0.7 (Galois Fields as 2-Adic Residue Fields). For each $m \geq 1$, the Galois field $GF(2^m) = \mathbb{F}_{2^m}$ used in the CRT co-processors of ADDONs 7–8 is the residue field of the unique (up to isomorphism) unramified extension of $\mathbb{Z}_2$ of degree m :

$$\mathbb{Z}_2^{(m)} / 2\mathbb{Z}_2^{(m)} \cong GF(2^m). \quad (6)$$

Consequently, the full algebraic stack of ADAM-PentaD

$$GF(2^m) \hookrightarrow \mathbb{Z}_2^{(m)} \hookrightarrow \mathbb{Q}_2$$

is a single coherent tower: every Galois arithmetic operation in the CRT layer lifts canonically to an exact 2-adic computation via Witt-vector arithmetic, without any change to the physical co-processor silicon.

Proof. The unramified extension $\mathbb{Z}_2^{(m)}$ is constructed by adjoining a root of an irreducible polynomial $f_m \in \mathbb{Z}_2[x]$ of degree m such that $f_m \bmod 2$ is the chosen irreducible in $\mathbb{F}_2[x]$ used by the CRT compiler (ADDON 7, Definition 1). The residue map $\mathbb{Z}_2^{(m)} \rightarrow GF(2^m)$ sends $x \mapsto x \bmod 2$ and is a surjective ring homomorphism with kernel $2\mathbb{Z}_2^{(m)}$. □ □

Remark 0.8 (Hensel’s Lemma as the p -Adic Kalman Lift). Hensel’s lemma (Hensel, 1897) states: if $f \in \mathbb{Z}_p[x]$ and $a_0 \in \mathbb{Z}_p$ satisfies $f(a_0) \equiv 0 \pmod{p}$ and $f'(a_0) \not\equiv 0 \pmod{p}$, then there exists a unique $a \in \mathbb{Z}_p$ with $f(a) = 0$ and $a \equiv a_0 \pmod{p}$. This is the non-Archimedean analogue of Newton’s method. In the ADAM-PentaD context, the Hybrid-Domain Photonic Kalman Filter (ADDON 5, Definition 2) performs the *same operation in the*

Archimedean domain: given a coarse estimate $\hat{z}_k^-$ (the “mod p ” approximation) and a correction step driven by the innovation residual, it lifts the estimate to a more precise posterior $\hat{z}_k$ (the “mod p^2 ” lift). The p-adic PCM substrate natively executes this lifting as a tree-descent, unifying the Kalman and Hensel frameworks in a single hardware pass.

11.30 p-Adic Wavelets and Connection to the Krylov-Lanczos Cascade

Definition 0.9 (Kozyrev p-Adic Wavelets). The *Kozyrev wavelet basis* of $L^2(\mathbb{Q}_p)$ (Kozyrev, 2002) consists of functions

$$\psi_{n,j,a}(x) := p^{-n/2} \chi(p^{-n-1} j x) \cdot \mathbf{1}_{a+p^n \mathbb{Z}_p}(x), \quad (7)$$

where $n \in \mathbb{Z}$ is the *scale*, $j \in \{1, \dots, p-1\}$ the *oscillation index*, $a \in \mathbb{Q}_p/p^n \mathbb{Z}_p$ the *translation*, and $\chi : \mathbb{Q}_p \rightarrow S^1$ is the additive character $\chi(x) = e^{2\pi i \{x\}_p}$ with $\{x\}_p$ the fractional part. The set $\{\psi_{n,j,a}\}$ is an orthonormal basis of $L^2(\mathbb{Q}_p)$.

Proposition 0.10 (Kozyrev Scales as Krylov Taps). *Each scale level n of the Kozyrev basis (Definition 0.9) corresponds to exactly one spatial tap $k = n$ of the Krylov subspace cascade (ADDON 6, Proposition 1). The tap amplitude $\mathbf{A}^k v_0$ is the projection of the wavefront onto the n -th-scale Kozyrev subspace:*

$$\mathbf{A}^k v_0 \approx \sum_{j,a} \langle v_0, \psi_{k,j,a} \rangle \psi_{k,j,a}, \quad (8)$$

with the approximation becoming exact as $m \rightarrow \infty$ in $\mathcal{K}_m(\mathbf{A}, v_0)$. The p-adic multi-resolution analysis thus provides the functional-analytic foundation that was assumed but not constructed in ADDON 6.

Proof sketch. The Vladimirov pseudo-differential operator D^α on $\mathbb{Q}_p$ has the Kozyrev wavelets as its eigenfunctions with eigenvalues $p^{(n+1)\alpha}$ (Kozyrev, 2002). If $\mathbf{A}$ is realised as a truncated Vladimirov operator on $\mathbb{Z}_p/p^d \mathbb{Z}_p$, then $\mathbf{A}^k$ rescales each Kozyrev mode by $p^{(n+1)k\alpha}$, reproducing the Krylov power-iteration structure tap by tap. $\square$

11.31 HoTT h-Level — p-Adic Valuation Correspondence

Proposition 0.11 (HoTT h-Levels and p-Adic Valuation). *Define the shifted valuation $\tilde{v}_p(x) := v_p(x) - 2$ for $x \in \mathbb{Z}_p \setminus \{0\}$. The following dictionary is a bijection of hierarchical structures:*

$$\begin{array}{ll} \text{h-level } -2 \longleftrightarrow \tilde{v}_p = 0 & \text{contractible type} \longleftrightarrow v_p(x) = 2 \\ \text{h-level } -1 \longleftrightarrow \tilde{v}_p = 1 & \text{proposition} \longleftrightarrow v_p(x) = 3 \\ \text{h-level } 0 \longleftrightarrow \tilde{v}_p = 2 & \text{set (discrete type)} \longleftrightarrow v_p(x) = 4 \\ \text{h-level } n \longleftrightarrow \tilde{v}_p = n + 2 & \text{n-groupoid} \longleftrightarrow v_p(x) = n + 4 \end{array} \quad (9)$$

Under this identification, the HoTT truncation map $\|\cdot\|_n$ (projecting a type to its n -truncation) corresponds to the p-adic truncation $x \mapsto x \bmod p^{n+4}$ in $\mathbb{Z}_p$.

Proof sketch. Both structures are complete, linearly ordered refinement towers: the h-level measures “how many non-trivial homotopy groups a type has,” and the p-adic valuation measures “at which power of p the element becomes non-trivial.” The truncation operations (homotopy truncation and p-adic truncation) are both universal in their respective categories: $\|A\|_n$ is the initial n -type receiving a map from A , and $x \bmod p^k$ is the initial quotient of $\mathbb{Z}_p$ resolving at level k . The bijection (9) preserves both the ordering and the universal property. $\square$

Remark 0.12 (Physical Interpretation: Semantic Depth as Address Depth). Proposition 0.11 closes the 3+1 convergence of ADDON 9 in hardware: the h-level of a HoTT type (Definitions 2–3 of ADDON 9) is no longer an abstract integer but the *physical depth* of the corresponding PCM node in the p-adic array (Definition 0.4). Propositions (h-level -1 , $v_p = 3$) live three levels from the root; sets (h-level 0) live four levels deep; and so on. The type checker of the Semantic Homotopy Filter (ADDON 9, Algorithm 1) translates to a tree-descent of depth $v_p(x) + 2$, executed natively by the PCM wiring without any software layer.

Algorithm 1 formalises context retrieval under p-adic addressing.

Remark 0.13 (Convergence of the Full Ten-ADDON Stack). ADDON 10 closes the mathematical architecture as a vertically integrated tower (Table 1).

Every structure introduced across ADDONs 4–10 lives inside a single algebraic tower anchored at the Clifford algebra $\mathbb{C}l(0, d)$: the Galois fields are its residue fields, the p-adic integers are their canonical lifts, the Kozyrev wavelets are the eigenbasis of the resulting pseudo-differential operator, and the HoTT type hierarchy is isomorphic to the p-adic valuation tower. No external axioms are required; the architecture is a single coherent mathematical object.

Algorithm 1: p -Adic PCM Context Lookup (per attention query q)

Input: Query $q \in \mathbb{Cl}(0, d)$; PCM tree of depth D ; Prüfer-code encoder $\phi : \mathbb{Cl}(0, d) \rightarrow \mathbb{Z}_p/p^D\mathbb{Z}_p$.

Output: Retrieved weight tensor W_q from the leaf matching q .

```
// --- Encode query as p-adic address ---
1  $\alpha \leftarrow \phi(q)$  // Prüfer code:  $D$ -digit base- $p$  integer;
// --- Descend tree:  $D$  steps, no TLB, no cache miss ---
2  $\text{node} \leftarrow \text{root}$ ;
3 for  $k \leftarrow D - 1$  to 0 do
4    $b_k \leftarrow \lfloor \alpha/p^k \rfloor \bmod p$  //  $k$ -th digit of address;
5    $\text{node} \leftarrow \text{child}(\text{node}, b_k)$  // 1 PCM cell read;
6   Apply Kalman correction from internal node (ADDON 5, Eqs. 5-9);
7  $W_q \leftarrow \text{leaf-weight}(\text{node})$ ;
8 return  $W_q$ ; // Total cost:  $D = O(\log_p N)$  cell reads
```

Layer	ADDONs	Domain	Key structure
Physical	4-5	Photonic substrate	Lyapunov, Kalman, $\mathcal{N}$
Algebraic	6-8	Galois / braid groups	CRT, B_3 , Landauer
Semantic	9	HoTT / Lambek	Types, paths, grammar
Memory	10	p -adic / PCM	$\mathbb{Q}_p$, Witt, Kozyrev
<i>Unifying algebra throughout: $\mathbb{Cl}(0, d)$, with $GF(2^m) \hookrightarrow \mathbb{Z}_2^{(m)} \hookrightarrow \mathbb{Q}_2$</i>			

Table 1: The ten-ADDON ADAM-PentaD mathematical stack.

«Воевать не числом, а умением».

“Wage war not by numbers, but by skill.”

A. V. Suvorov,

Наука побеждать / The Science of Victory, c. 1799